Lecture 16 (01/08/20)

Eigenvector and Eigenvalue

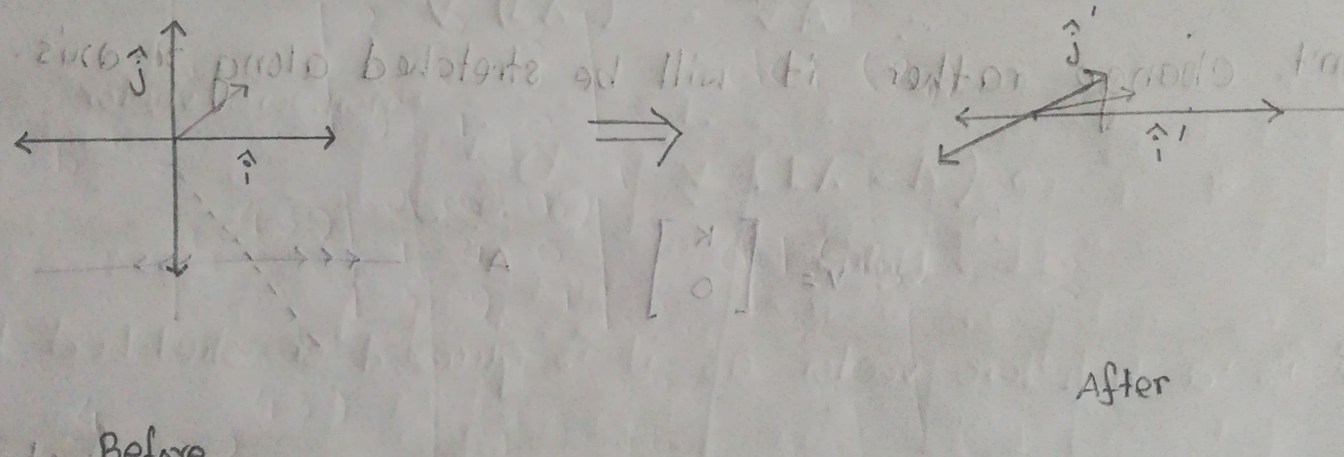
To understand this lecture

① Linear Transformation

② Determinants

③ Change of Basis

Linear transformation done by the matrix,
$$A = \begin{array}{c} \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \\ \hat{i} \quad \hat{j} \end{array}$$



A vector will be deviated from its span (straight line) if the basis are changed.

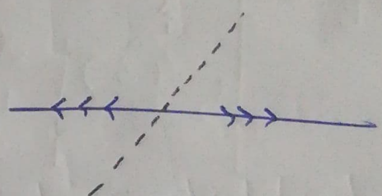
Some vectors will remain in it's span even after being multiplied by that matrix. Such as any vector in x axis.

$v = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ $\qquad$ $Av = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 3 \cdot \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

(unchanged span → stretched along x-axis)

If a vector $\begin{bmatrix} k \\ 0 \end{bmatrix}$ is multiplied with A, the span won't change rather it will be stretched along x-axis.

$v = \begin{bmatrix} k \\ 0 \end{bmatrix}$ $\qquad$ A



If a vector $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ it will stretched by some factor but it's span wont change

$v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $\qquad$ $Av = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Eigenvectors:

The vectors which stays on the same span after transformation.

Same span → stretched / squeezed.

Eigenvalue:

The factor by which the eigenvectors get squeezed or stretched.

Eigenvalues have very strong relationship with determinants of a matrix.

Eigenvalues can be negative

* If the eigenvalue is $\boxed{-\frac{1}{2}}$, it means the orientation of the basis vectors will be flipped.

-ve value signifies that it has something to do with orientation.

$$\underbrace{A \vec{v}}_{\text{Eigen-vector}} = \underbrace{\lambda \vec{v}}_{\text{Eigen-value}}$$

But there's inconsistency in the eq^n.

L.H.S = matrix * vector

R.H.S = scalar * vector

$$\therefore A\vec{v} = (\lambda I)\,\vec{v}$$

non.zero vector

$$\Rightarrow \underbrace{(A - \lambda I)}_{\text{Matrix}}\,\vec{v} = 0$$

* $\vec{v}$ is a non-zero vector as it is queezed or stretched by A.
* $A - \lambda I$ is such a matrix which squeezes the space of the non.zero vector $\vec{v}$ into a lower dimension. (i.e straight line → point)

$\therefore \det(A - \lambda I) = 0$ → The factor by which $\vec{v}$ is squeezed = 0.

☐ How to solve $Ax = \lambda x$?

$$(A - \lambda I)x = 0$$

$$\therefore \det(A - \lambda I) = 0$$

① Find λ first

② Then find the nullspace, which will be x.

<u>Example:</u>

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

① Symmetric
② Constant along diagonal

$$A - \lambda I = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{bmatrix}$$

* $\det(A - \lambda I) = 0$

$$\therefore \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = 0$$

$$\therefore \lambda = 2, 4$$

$$(A - \lambda I)\vec{x} = 0$$

For $\lambda = 2$

$$\begin{bmatrix} 3-2 & 1 \\ 1 & 3-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$x + y = 0$

$x + y = 0$

$x = -1, \; y = 1$

$$\therefore \vec{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

For $\lambda = 4$

$$\begin{bmatrix} 3-4 & 1 \\ 1 & 3-4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$\therefore x = 1, \; y = 1$

$$\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Check

$$Ax = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = \underline{\underline{4}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Eigen values

$$Ax = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \underline{\underline{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Example:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$ for λ, $\det(A - \lambda I) = 0$

$$\begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda = 1, -1$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} [x] = 0 \quad \therefore x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (\lambda = 1)$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} [x] = 0 \quad x = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad (\lambda = -1)$$

To get Eigenvectors and Eigenvalues

1) Compute the determinant of $(A - \lambda I)$

2) Find the roots of the polynomials of $\det(A - \lambda I) = 0$.

3) For each eigenvalue λ, solve $(A - \lambda I)x = 0$ to find eigen-vectors.

* no. of Eigenvectors = no. of Eigenvalues

* for n by n matrix, product of n eigenvalues is equal to the determinant.

Sum of n eigenvalues = Sum of n diagonal entries. (Trace)

Relationship between eigenvalue and determinants.

$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ det $(A) = 9 - 1 = 8$

Eigenvalues, $\lambda = 4, 2$ $\therefore \det(A) = 4 \times 2 = 8 = \text{pro}(\lambda_1, \lambda_2)$

Sum of diagonal entries, Trace $(A) = (3+3) = 6$

Sum of $n = 2$ eigenvalues, sum $(\lambda_1, \lambda_2) = 4 + 2 = 6$

$\square$ $Ax = \lambda x$

$$\Rightarrow (A + kI)x = Ax + kIx$$
$$= \lambda x + kIx$$
$$= (\lambda + kI)x$$

If 'kI' is added to any matrix, the eigenvalue is increased by k. (And eigen vector remains unchanged)

$A_1 = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ $\lambda = 2, 4$

$\uparrow 3 \quad \uparrow 3$ $A_2 + 3I = A_1$

$A_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\lambda = 1, -1$

There's a common misconception that, If

$Ax = \lambda x$ | where λ and α both are eigenvalues.

$Bx = \alpha x$

$$(A+B)x = (\alpha + \lambda)x \quad \times$$

This is only true when the eigenvectors are same for both the cases

Not true because the eigen vector x might not be same for A and B.

Example

Rotation matrix (90° rotation), $g = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Trace, $\lambda_1 + \lambda_2 = 0$

$\det(g) = \lambda_1 . \lambda_2 = 1$

$\lambda^2 + 1 = 0$ | $x^2 + (\alpha+\beta)x + \alpha\beta = 0$

$\Rightarrow \lambda = \pm\sqrt{-1} = (+i), (-i)$

* After we rotate a vector (with a rotation matrix), the vector can't be in its span anymore only if we are not in an imaginary space

So after being multiplied with a rotation matrix, no vector can remain in its own span. (unless it's in an imaginary space, then we can reverse the rotation and get the initial vector

" The more symmetric the matrix is, the better our eigen-values will be "

* Not every matrix will have multiple eigen-values.

Example:

$$\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$$

$\lambda_1 + \lambda_2 = 6$

$\lambda_1 \lambda_2 = 9$

triangular matrix

$\lambda^2 - 6\lambda + 9 = 0$

$\Rightarrow \lambda_1 = 3 \quad \lambda_2 = 3 \qquad \therefore \lambda_1 = \lambda_2 = 3$

$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad x_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\therefore$ only 1 independent eigenvector

$\therefore$ For triangular matrices, we won't get n eigen-values for $n \times n$ matrix.

No. of eigen value depends on
→ size of the given matrix
→ properties of " " (symmetry)

H.W

☐ Find Generalised characteristics of Eigenvalues and Eigenvectors of

1. Projection matrices
2. Reflection matrices
3. Rotation matrices

Projection Matrix

Let, P be a projection matrix,

$\therefore Px = \lambda x$

$\Rightarrow P^2 x = \lambda x$ [For projection matrix, $P = P^2$]

$\Rightarrow P.(Px) = \lambda x$ [$Px = \lambda x$]

$\Rightarrow P.\lambda x = \lambda x$

$\Rightarrow \lambda.(Px) = \lambda x$

$\Rightarrow \lambda^2 x - \lambda x = 0$

$\Rightarrow \lambda = 1, 0$ [$x \neq$ zero vector]

Let, the projection matrix of $\vec{a} = \begin{bmatrix} a \\ b \end{bmatrix}$ be P,

$$\therefore P = \frac{a.a^T}{a^T.a} = \frac{1}{a^2+b^2}\begin{bmatrix} a \\ b \end{bmatrix}\begin{bmatrix} a & b \end{bmatrix} = \frac{1}{a^2+b^2}\begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix}$$

For $\lambda = 1$,

$$(P - \lambda I) = \frac{1}{a^2+b^2}\begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \frac{1}{a^2+b^2}\begin{bmatrix} a^2-(a^2+b^2) & ab \\ ab & b^2-(a^2+b^2) \end{bmatrix}$$

$$= \frac{1}{a^2+b^2}\begin{bmatrix} -b^2 & ab \\ ab & -a^2 \end{bmatrix}$$

$\therefore$ Nullspace of the given vector, eigenvector $= \begin{bmatrix} a \\ b \end{bmatrix}$

For $\lambda = 0$, $(P - \lambda I) = \frac{1}{a^2+b^2}\begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix}$

$\therefore$ Eigenvector $= \begin{bmatrix} b \\ -a \end{bmatrix}$ $\quad (\perp \begin{bmatrix} a \\ b \end{bmatrix})$

Take aways

① Eigenvalues for projection matrix is always either $\underline{\underline{0}}$ or $\underline{\underline{1}}$.

② Eigenvector when $\underline{\lambda = 1}$, is the vector whose responsible for projection matrix P.

Eigenvector when $\underline{\lambda = 0}$, is the vector perpendicular to the vector for which we determined P.

$x_1 \cdot x_2 = 0$

Rotation Matrix

Let, rotation matrix, $R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

$\therefore$ $\text{Trace}(R) = \lambda_1 + \lambda_2 = 2\cos\theta$

$\text{Det}(R) = \lambda_1 \lambda_2 = \cos^2\theta + \sin^2\theta = 1$

$\therefore$ $\lambda^2 - 2\cos\theta\lambda + \cos^2\theta + \sin^2\theta = 0$

$\Rightarrow (\lambda - \cos\theta)^2 = -\sin^2\theta = (i\sin\theta)^2$

$\Rightarrow \lambda = \cos\theta \pm i\sin\theta$

For $\lambda = \cos\theta + i\sin\theta$,

$\therefore$ $(R - \lambda I) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} - \begin{bmatrix} \cos\theta + i\sin\theta & 0 \\ 0 & \cos\theta + i\sin\theta \end{bmatrix}$

$= \begin{bmatrix} -i\sin\theta & -\sin\theta \\ \sin\theta & -i\sin\theta \end{bmatrix} \times i$

$\therefore$ Eigenvector $= \begin{bmatrix} 1 \\ -i \end{bmatrix}$

$\therefore$

$\therefore$ For $\lambda = \cos\theta - i\sin\theta$

$$(R - \lambda I) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} - \begin{bmatrix} \cos\theta - i\sin\theta & 0 \\ 0 & \cos\theta - i\sin\theta \end{bmatrix}$$

$$= \begin{bmatrix} i\sin\theta & -\sin\theta \\ \sin\theta & i\sin\theta \end{bmatrix} \Big) * i$$

$\therefore$ Eigenvector $= \begin{bmatrix} 1 \\ i \end{bmatrix}$

<u>Takeaways</u>

① Eigenvalues appear as complex conjugate pairs, that is $a \pm ib$, $\cos\theta \pm i\sin\theta$

② Their respective eigenvectors are also complex conjugates, as we can see $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ i \end{bmatrix}$

③ As here rotation occurs, the vector can't exist in its previous span anymore, so we can notice that its new span is imaginary space.

Reflection matrix

For simplicity, let us consider, reflection matrix, $R = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

which reflects a matrix multiplied to it about $x = y$ line.

$\therefore$ Trace (R), $\lambda_1 + \lambda_2 = 0$

det (R), $\lambda_1 . \lambda_2 = -1$

$\therefore \lambda^2 - 1 = 0 \Rightarrow \lambda = 1, -1$ Eigenvalues,

For $\underline{\lambda = 1}$,

$R - \lambda I = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$\therefore$ Eigen-vector $= \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (This vector lies on the line $x = y$)

For $\lambda = -1$,

$R - \lambda I = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$\therefore$ Eigen-vector $= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

(which represents the reflection of a point perpendicular to $y = x$.)

Take aways -

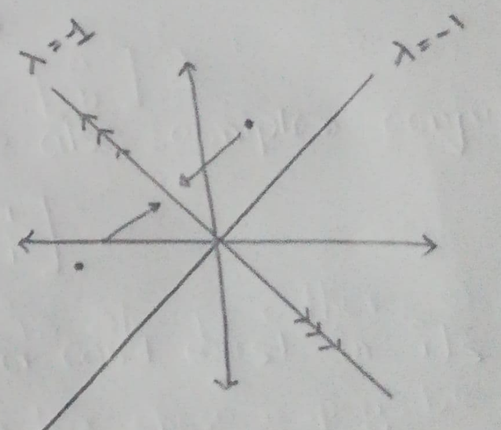
1. The eigenvalues of reflection matrix is either -1 or 1.

2. For $\lambda = 1$, the eigen vector that is obtained lies on the line about which the reflection occurs.

(The vector can be squeezed or stretched along that straight line / reference line of reflection)

3. For $\lambda = -1$, the obtained eigen-vector is the vector that will reflect with respect to the straight line / vector obtained due to $\lambda = 1$.

The vector due to $\lambda = 1$ is fixed, but for $\lambda = -1$, the points will be reflected along the vector due to $\lambda = 1$.



And, $x_1 \perp x_2$.